

# ❏ 1. Basic Service Security

## ❏ Overview

Service security ensures that individual services (like web, DNS, SSH, FTP) are:

- Protected from unauthorized access
- Properly configured
- Patched and updated

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## ❏ Key Principles:

| Principle       | Description                                     |
|-----------------|-------------------------------------------------|
| Least Privilege | Services run with minimum necessary permissions |

|                        |                                                    |
|------------------------|----------------------------------------------------|
| Service Isolation      | Use of containers/chroot/jails to isolate services |
| Minimal Installation   | Install only required services                     |
| Firewall Configuration | Block unnecessary ports                            |
| Patch Management       | Regular updates to remove vulnerabilities          |

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## ☒ Secure Service Configuration Steps:

1. **Disable Unused Services:**

bash

Copy code

```
systemctl disable telnet
```

2.

**Limit Access with Firewalls (iptables/nftables):**

bash

Copy code

```
iptables -A INPUT -p tcp --dport 22 -s 192.168.1.0/24 -j ACCEPT
```

3.

**Use Secure Protocols:**

- Prefer SSH over telnet
- Use HTTPS over HTTP

4.

**Log Monitoring:**

- Monitor logs in /var/log/
- Use auditd, fail2ban for suspicious activity

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## ❏ 2. Logging and NTP (Network Time Protocol)

### ❏ Logging:

| Tool       | Purpose                          |
|------------|----------------------------------|
| rsyslog    | Modern log daemon for Linux      |
| journalctl | Systemd log viewer               |
| logrotate  | Manages log size and rotation    |
| auditd     | Records security-relevant events |

## ❏ Important Log Files:

| Path              | Contains                |
|-------------------|-------------------------|
| /var/log/messages | System-wide logs        |
| /var/log/auth.log | Authentication logs     |
| /var/log/syslog   | General system messages |
| /var/log/secure   | RedHat-based auth logs  |

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## ❏ Best Practices for Logs:

- - Store logs on separate partition
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Encrypt sensitive logs

- Use centralized log server (e.g., syslog-ng)
- Enable logrotate to manage old logs

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## ☒ Network Time Protocol (NTP):

- Ensures **accurate system time**, important for:
  - Log correlation
  - Security tokens/tickets
  - Cron jobs

## ☒ Common NTP tools:

bash

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timedatectl status    # Show time sync status ntpq -p            # Query NTP servers chronyc sources            # Chrony tool

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## ☒ 3. BIND and DNS Security

### ☒ What is BIND?

- BIND (Berkeley Internet Name Domain) is the most widely used DNS server on the internet.
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Default config file: /etc/named.conf

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## ⌘ DNS Security Threats:

| Threat                | Description                              |
|-----------------------|------------------------------------------|
| DNS Spoofing          | Fake DNS replies from attackers          |
| Cache Poisoning       | Corrupting DNS cache with malicious data |
| Zone Transfer Attacks | Gaining access to entire DNS database    |
| Amplification Attacks | Used in DDoS via large UDP DNS responses |

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## ⌘ BIND Security Practices:

1.



### **Disable Zone Transfers:**

c

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```
allow-transfer { none; };
```

2.

### **Use Access Controls:**

conf

Copy code

```
allow-query { localhost; };
```

3.

### **Run BIND in Chroot Jail:**

```
bash
```

Copy code

```
yum install bind-chroot
```

4.

**Enable DNSSEC (DNS Security Extensions):**

- Verifies DNS records with digital signatures

5.

**Use logging in BIND:**

```
conf
```

Copy code

```
logging { channel default_log { file "/var/log/named.log"; severity info; }; }
```

6.

**Keep software updated** to patch known vulnerabilities.